

- 8. Suppose that $A = \{2, 4, 6\}$, $B = \{2, 6\}$, $C = \{4, 6\}$, and $D = \{4, 6, 8\}$. Determine which of these sets are subsets of which other of these sets.

Answer:

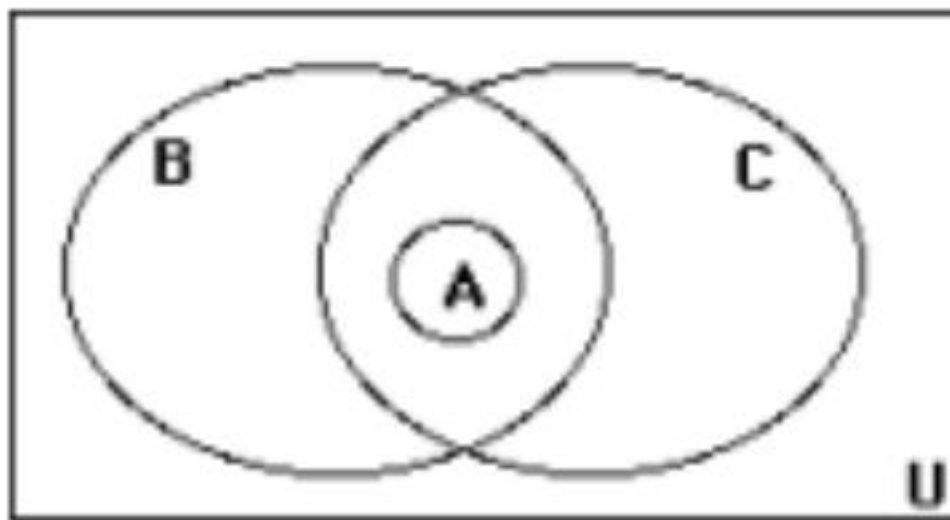
- Each of the sets is a subset of itself (易忽略) .
- Aside from that, the only relations are $B \subseteq A$, $C \subseteq A$, and $C \subseteq D$.

- 12 Determine whether these statements are true or false.
- f) $\{\{\emptyset\}\} \subset \{\emptyset, \{\emptyset\}\}$
- g) $\{\{\emptyset\}\} \subset \{\{\emptyset\}, \{\emptyset\}\}$

正确答案：

- f) true
- g) false

- Use a Venn diagram to illustrate the relationships $A \subset B$ and $A \subset C$.
- 正确答案（**B和C的关系题目未给出，不能轻易下定论**）：



- 24 Can you conclude that $A = B$ if A and B are two sets with the same power set?
- 参考答案：
- The union of all the sets in the power set of a set X must be exactly X . In other words, we can recover X from its power set, uniquely. Therefore the answer is yes.

- 48 Find the truth set of each of these predicates where the domain is the set of **integers**.
- 常见错误：集合中没有注明 $x \in Z$
- 正确答案（若能求解应化简至最终结果）：
- (a) $P(x): x^3 \geq 1$ 答案： $\{x \in Z \mid x^3 \geq 1\} = \{x \in Z \mid x \geq 1\} = \{1, 2, 3, \dots\}$
- (b) $Q(x): x^2 = 2$ $\emptyset$ 或 $\{\}$
- (c) $R(x): x < x^2$ $\{x \in Z \mid x < x^2\} = \{x \in Z \mid x < 0 \vee x > 1\} = \{\dots, -3, -2, -1, 2, 3, \dots\}$
- 答案表示方法不唯一

2.3 20题

Give an example of a function from $\mathbb{N}$ to $\mathbb{N}$ that is

- a) one-to-one but not onto
- b) onto but not one-to-one
- c) both onto and one-to-one (but different from the identity function)
- d) neither one-to-one nor onto

参考答案:

- a) $f(n) = n+19$
- b) $f(n) = \lfloor n/4 \rfloor$ (结果取整)
- c) $f(n) = n+1$ for n is even, $f(n) = n-1$ for n is odd.
- d) $f(n) = 19$

2.3 33题

Suppose that g is a function from A to B and f is a function from B to C .

- a) Show that if both f and g are one-to-one functions, then $f \circ g$ is also one-to-one
- b) Show that if both f and g are onto functions, then $f \circ g$ is also onto

参考答案:

a) Let x and y be different values of A , since g is one-to-one, $g(x) \neq g(y)$ we have $f(g(x)) \neq f(g(y))$, hence, $f \circ g$ is one-to-one.

b) Let $y \in C$, because f is onto, $y = f(b)$ for $b \in B$, and we have $b = g(x)$ for $x \in A$, hence, $y = f(b) = f(g(x)) = f \circ g$, $f \circ g$ is onto.

2.4

6. List the first 10 terms of each of these sequences.

(g) the sequence whose terms are constructed sequentially as follows: start with 1, then add 1, then multiply by 1, then add 2, then multiply by 2, and so on.

参考答案:

g) 1, 2, 2, 4, 8, 11, 33, 37, 148, 153

2.4

26. For each of these lists of integers, provide a simple formula or rule that generates the terms of an integer sequence that begins with the given list.

Assuming that your formula or rule is correct, determine the next three terms of the sequence.

f) 1, 3, 15, 105, 945, 10395, 135135, 2027025, 34459425, ...

参考答案:

f) We notice that each term evenly divides the next, and the multipliers are successively 3, 5, 7, 9, 11, and so on. That must be the intended pattern. One notation for this is to use $n!!$ to mean $n(n-2)(n-4)\cdots$; thus the n^{th} term is $(2n-1)!!$. Thus the next three terms are 654729075, 13749310575, 316234143225.

30. What are the values of these sums, where $S = \{1, 3, 5, 7\}$?

b) $\sum_{j \in S} j^2$

d) $\sum_{j \in S} 1$

Answer :

b) $1^2 + 3^2 + 5^2 + 7^2 = 84$

d) $1 + 1 + 1 + 1 = 4$

常见问题：许多同学答案有误

34. Compute each of these double sums.

b) $\sum_{i=0}^3 \sum_{j=0}^2 (3i + 2j)$

Answer :

b) $(0+0) + (0+2) + (0+4) + (3+0) + (3+2) + (3+4) + (6+0) + (6+2) + (6+4) + (9+0) + (9+2) + (9+4) = 78$

常见问题：许多同学答案有误

- 2.5 2. Determine whether each of these sets is finite, countably infinite, or uncountable. For those that are countably infinite, exhibit a one-to-one correspondence between the set of positive integers and that set.
- d) the real numbers between 0 and 2
- e) the set $A \times \mathbb{Z}^+$ where $A = \{2, 3\}$
- f) the integers that are multiples of 10
- 参考答案:
- d) uncountable
- e) This set is countable. We can list its elements in the order $(2,1), (3,1), (2,2), (3,2), (2,3), (3,3), \dots$, giving us the one-to-one correspondence $1 \leftrightarrow (2,1), 2 \leftrightarrow (3,1), 3 \leftrightarrow (2,2), 4 \leftrightarrow (3,2), 5 \leftrightarrow (2,3), 6 \leftrightarrow (3,3), \dots$
- f) This set is countable. The integers in the set are $0, \pm 10, \pm 20, \pm 30$, and so on. We can list these numbers in the order $0, 10, -10, 20, -20, 30, \dots$, thereby establishing the desired correspondence. In other words, the correspondence is given by $1 \leftrightarrow 0, 2 \leftrightarrow 10, 3 \leftrightarrow -10, 4 \leftrightarrow 20, 5 \leftrightarrow -20, 6 \leftrightarrow 30$, and so on.

- 2.5 10. Give an example of two uncountable sets A and B such that $A - B$ is
 - a) finite.
 - b) countably infinite.
 - c) uncountable.
 - 参考答案:
- In each case, let us take A to be the set of real numbers.
- a) We can let B be the set of real numbers as well; then $A - B = \emptyset$, which is finite.
- b) We can let B be the set of real numbers that are not positive integers; in symbols, $B = A - \mathbb{Z}^+$. Then $A - B = \mathbb{Z}^+$, which is countably infinite.
- c) We can let B be the set of positive real numbers. Then $A - B$ is the set of negative real numbers and 0, which is certainly uncountable.

4. Find the product AB , where

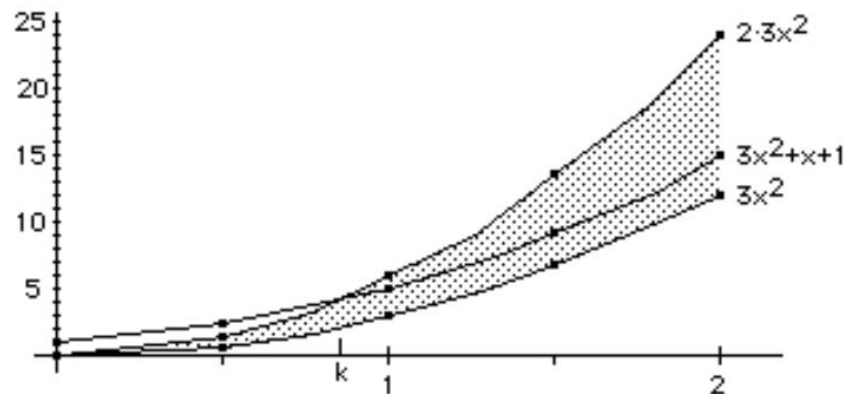
$$\text{b) } \mathbf{A} = \begin{bmatrix} 1 & -3 & 0 \\ 1 & 2 & 2 \\ 2 & 1 & -1 \end{bmatrix}, \mathbf{B} = \begin{bmatrix} 1 & -1 & 2 & 3 \\ -1 & 0 & 3 & -1 \\ -3 & -2 & 0 & 2 \end{bmatrix}.$$

Answer :

$$\text{b) } \begin{bmatrix} 4 & -1 & -7 & 6 \\ -7 & -5 & 8 & 5 \\ 4 & 0 & 7 & 3 \end{bmatrix}$$

常见问题：部分同学答案有误

- 3.2 34.
- a) Show that $3x^2+x+1$ is $\theta(3x^2)$ by directly finding the constants k , C_1 , and C_2 .
- b) Express the relationship in part (a) using a picture showing the functions $3x^2+x+1$, $C_1 \cdot 3x^2$, and $C_2 \cdot 3x^2$, and the constant k on the x-axis, where C_1 , C_2 , and k are the constants you found in part (a) to show that $3x^2+x+1$ is $\theta(3x^2)$.
- 参考答案: a) By Exercise 31 we have to show that $3x^2+x+1$ is $O(3x^2)$ and that $3x^2$ is $O(3x^2+x+1)$. The latter is trivial, since $3x^2 \leq 3x^2+x+1$ for $x > 0$. The former is almost as trivial, since $3x^2+x+1 \leq 3x^2+3x^2 = 2 \cdot 3x^2$ for all $x > 1$. What we have shown is that $1 \cdot 3x^2 \leq 3x^2+x+1 \leq 2 \cdot 3x^2$ for all $x > 1$; in other words, $C_1 = 1$ and $C_2 = 2$ in Exercise 33.
- b) The following picture shows that graph of $3x^2+x+1$ falls in the shaded region between the graph of $3x^2$ and the graph of $2 \cdot 3x^2$ for all $x > 1$.



- 3.2 74. Determine whether $\log n!$ is $\Theta(n \log n)$. Justify your answer.
- 参考答案:

A big- O estimate for $n!$ can be obtained by noting that each term in the product does not exceed n . Hence, $n! = 1 \cdot 2 \cdot 3 \cdots n \leq n \cdot n \cdot n \cdots n = n^n$. This inequality shows that $n!$ is $O(n^n)$, taking $C = 1$ and $k = 1$ as witnesses. Taking logarithms of both sides of the inequality established for $n!$, we obtain $\log n! \leq \log n^n = n \log n$. This implies that $\log n!$ is $O(n \log n)$.

We can easily show that $(n-i)(i+1) \geq n$ for $i = 0, 1, \dots, n-1$. Hence, $(n!)^2 = (n \cdot 1)((n-1) \cdot 2) \cdot ((n-2) \cdot 3) \cdots (2 \cdot (n-1))(1 \cdot n) \geq n^n$. Therefore, $2 \log n! \geq n \log n$.

In conclusion, $\log n!$ is $\Theta(n \log n)$.

Section 4.2

26. Use Algorithm 5 to find $11^{644} \bmod 645$.

Initial $x=1$

Initial power: $b^{2^0}=11^{2^0} \bmod 645=11$

$$644 = (1010000100)_2$$

$i = 0$: Because $a_0 = 0$, we have $x = 1$ and $power = 11^2 \bmod 645 = 121 \bmod 645 = 121$;

$i = 1$: Because $a_1 = 0$, we have $x = 1$ and $power = 121^2 \bmod 645 = 14641 \bmod 645 = 451$;

$i = 2$: Because $a_2 = 1$, we have $x = 1 \cdot 451 \bmod 645 = 451$ and $power = 451^2 \bmod 645 = 203401 \bmod 645 = 226$;

$i = 3$: Because $a_3 = 0$, we have $x = 451$ and $power = 226^2 \bmod 645 = 51076 \bmod 645 = 121$;

$i = 4$: Because $a_4 = 0$, we have $x = 451$ and $power = 121^2 \bmod 645 = 14641 \bmod 645 = 451$;

$i = 5$: Because $a_5 = 0$, we have $x = 451$ and $power = 451^2 \bmod 645 = 203401 \bmod 645 = 226$;

$i = 6$: Because $a_6 = 0$, we have $x = 451$ and $power = 226^2 \bmod 645 = 51076 \bmod 645 = 121$;

$i = 7$: Because $a_7 = 1$, we find that $x = (451 \cdot 121) \bmod 645 = 391$ and $power = 121^2 \bmod 645 = 14641$

$\bmod 645 = 451$;

$i = 8$: Because $a_8 = 0$, we have $x = 391$ and $power = 451^2 \bmod 645 = 203401 \bmod 645 = 226$;

$i = 9$: Because $a_9 = 1$, we find that $x = (391 \cdot 226) \bmod 645 = 1$.

16. Determine whether the integers in each of these sets are pairwise relatively prime.

a) 21, 34, 55

b) 14, 17, 85

c) 25, 41, 49, 64

d) 17, 18, 19, 23.

Solution:

Since these numbers are small, the easiest approach is to find the prime factorization of each number and look for any common prime factors.

a) Since $21 = 3 \cdot 7$, $34 = 2 \cdot 17$, and $55 = 5 \cdot 11$, these **are** pairwise relatively prime.

b) Since $85 = 5 \cdot 17$, these **are not** pairwise relatively prime.

c) Since $25 = 5^2$, 41 is prime, $49 = 7^2$, and $64 = 2^6$, these **are** pairwise relatively prime.

d) Since 17, 19, and 23 are prime and $18 = 2 \cdot 3^2$, these **are** pairwise relatively prime.

常见问题：部分同学a和c错误

24. What are the greatest common divisors of these pairs of integers?

a) $2^2 \cdot 3^3 \cdot 5^5, 2^5 \cdot 3^3 \cdot 5^2$

c) $17, 17^{17}$

f) $2 \cdot 3 \cdot 5 \cdot 7, 2 \cdot 3 \cdot 5 \cdot 7$

Solution:

We form the greatest common divisors by finding the minimum exponent for each prime factor.

a) $2^2 \cdot 3^3 \cdot 5^2$ c) 17 f) $2 \cdot 3 \cdot 5 \cdot 7$

常见问题：很多同学c) 错误，写成 17^{17}

40. Using the method followed in Example 17, express the greatest common divisor of each of these pairs of integers as a linear combination of these integers.

a) 9, 11 b) 33, 44 c) 35, 78

d) 21, 55 e) 101, 203 f) 124, 323

g) 2002, 2339 h) 3457, 4669 i) 10001, 13422.

常见问题：部分同学i) 题有误

$$13422 = 10001 + 3421$$

$$10001 = 2 \cdot 3421 + 3159$$

$$3421 = 3159 + 262$$

$$3159 = 12 \cdot 262 + 15$$

$$262 = 17 \cdot 15 + 7$$

$$15 = 2 \cdot 7 + 1$$

Thus the greatest common divisor is 1.

$$1 = 15 - 2 \cdot 7$$

$$1 = 15 - 2 \cdot (262 - 17 \cdot 15)$$

$$= 35 \cdot 15 - 2 \cdot 262$$

$$= 35 \cdot (3159 - 12 \cdot 262) - 2 \cdot 262$$

$$= 35 \cdot 3159 - 422 \cdot 262$$

$$= 35 \cdot 3159 - 422 \cdot (3421 - 3159)$$

$$= 457 \cdot 3159 - 422 \cdot 3421$$

$$= 457 \cdot (10001 - 2 \cdot 3421) - 422 \cdot 3421$$

$$= 457 \cdot 10001 - 1336 \cdot 3421$$

$$= 457 \cdot 10001 - 1336 \cdot (13422 - 10001)$$

$$= 1793 \cdot 10001 - 1336 \cdot 13422$$

6. Find an inverse of a modulo m for each of these pairs of relatively prime integers using the method followed in Example 2.

a) $a = 2, m = 17$ b) $a = 34, m = 89$

c) $a = 144, m = 233$ d) $a = 200, m = 1001$.

Solution:

a) $17 = 8 \cdot 2 + 1$, which means that $17 - 8 \cdot 2 = 1$, so -8 is an inverse. We can also report this as 9, because $-8 \equiv 9 \pmod{17}$.

b) We need to find s and t such that $34s + 89t = 1$. Then s will be the desired inverse, since $34s \equiv 1 \pmod{89}$. First we go through the Euclidean algorithm computation that $\gcd(34, 89) = 1$:

$$89 = 2 \cdot 34 + 21 \quad 34 = 21 + 13 \quad 21 = 13 + 8 \quad 13 = 8 + 5 \quad 8 = 5 + 3 \quad 5 = 3 + 2 \quad 3 = 2 + 1$$

Then we reverse our steps and write 1 as the desired linear combination:

$$\begin{aligned} 1 &= 3 - 2 \\ &= 3 - (5 - 3) = 2 \cdot 3 - 5 \\ &= 2 \cdot (8 - 5) - 5 = 2 \cdot 8 - 3 \cdot 5 \\ &= 2 \cdot 8 - 3 \cdot (13 - 8) = 5 \cdot 8 - 3 \cdot 13 \\ &= 5 \cdot (21 - 13) - 3 \cdot 13 = 5 \cdot 21 - 8 \cdot 13 \\ &= 5 \cdot 21 - 8 \cdot (34 - 21) = 13 \cdot 21 - 8 \cdot 34 \\ &= 13 \cdot (89 - 2 \cdot 34) - 8 \cdot 34 = 13 \cdot 89 - 34 \cdot 34 \end{aligned}$$

Thus $s \equiv -34$, so an inverse of 34 modulo 89 is -34 , which can also be written as 55.

12. Solve each of these congruences using the modular inverses found in parts (b), (c), and (d) of Exercise 6.

a) $34x \equiv 77 \pmod{89}$

b) $144x \equiv 4 \pmod{233}$

c) $200x \equiv 13 \pmod{1001}$.

Solution:

In each case we multiply both sides of the congruence by the inverse found in Exercise 6 and simplify. Our answers are not unique, of course—anything in the same congruence class works just as well.

a) We found that 55 is an inverse of 34 modulo 89, so $x \equiv 77 \cdot 55 = 4235 \equiv 52 \pmod{89}$. Check: $34 \cdot 52 = 1768 \equiv 77 \pmod{89}$.

b) We found that 89 is an inverse of 144 modulo 233, so $x \equiv 4 \cdot 89 = 356 \equiv 123 \pmod{233}$. Check: $144 \cdot 123 = 17712 \equiv 4 \pmod{233}$.

c) We found that -5 is an inverse of 200 modulo 1001, so $x \equiv 13 \cdot (-5) = -65 \equiv 936 \pmod{1001}$. (We could also leave the answer as -65 .) Check: $200 \cdot 936 = 187200 \equiv 13 \pmod{1001}$

5.1 54. Use mathematical induction to show that given a set of $n+1$ positive integers, none exceeding $2n$, there is at least one integer in this set that divides another integer in the set.

- The base case is $n = 1$. a set of two elements from $\{1, 2\}$.
- Assume the inductive hypothesis $p(n)$: among any $n+1$ positive integers not exceeding $2n$ there must be an integer that divides one of the other integers.
- consider a set A of $n + 2$ elements from $\{1, 2, \dots, 2n, 2n + 1, 2n+2\}$.
 - If $n+1$ of the elements of A are less than $2n + 1$, we get the conclusion.
 - assume that both $2n + 1$ and $2n + 2$ are in A , together with n smaller elements.
 - If $n + 1 \in A$, then we are done, since $n + 1 \mid 2n + 2$.
 - So we can assume that $n + 1 \notin A$.
 - $B = A - \{2n + 1, 2n + 2\} \cup \{n + 1\}$.
 - Since B is a collection of $n + 1$ numbers from $\{1, 2, \dots, 2n\}$, one element of B divides another.
 - If $n + 1$ is not one of these two numbers, then we are done.
 - assume $n + 1$ is one of these two numbers. Certainly $n + 1$ can't be the divisor, since its smallest multiple is too big to be in B , so there is some $k \in B$ that divides $n + 1$. But now k and $2n + 2$ are numbers in A , with k dividing $n + 2$, and we are done.

4. Let $P(n)$ be the statement that a postage of n cents can be formed using just 4-cent stamps and 7-cent stamps. The parts of this exercise outline a strong induction proof that $P(n)$ is true for $n \geq 18$.
- a) Show statements $P(18)$, $P(19)$, $P(20)$, and $P(21)$ are true, completing the basis step of the proof.
 - b) What is the inductive hypothesis of the proof?
 - c) What do you need to prove in the inductive step?
 - d) Complete the inductive step for $k \geq 21$.
 - e) Explain why these steps show that this statement is true whenever $n \geq 18$.

Solution:

- a) $P(18)$ is true, $18 = 1(4) + 2(7)$. $P(19)$ is true, $19 = 3(4) + 1(7)$. $P(20)$ is true, $20 = 5(4)$. $P(21)$ is true, $21 = 3(7)$.
- b) The inductive hypothesis $p(k)$: using just 4-cent and 7-cent stamps we can form j cents postage for all j with $18 \leq j \leq k$, where we assume that $k \geq 21$.
- c) In the inductive step we must show $p(k+1)$, assuming the inductive hypothesis, that we can form $k + 1$ cents postage using just 4-cent and 7-cent stamps.
- d) We want to form $k + 1$ cents of postage. Since $k \geq 21$, we know that $P(k - 3)$ is true, that is, that we can form $k - 3$ cents of postage. Put one more 4-cent stamp on the envelope, and we have formed $k + 1$ cents of postage, as desired.
- e) We have completed both the basis step and the inductive step, so by the principle of strong induction, the statement is true for every integer n greater than or equal to 18.

5. 3 50. Find these values of Ackermann's function.

a) $A(1, 0)$

b) $A(0, 1)$

c) $A(1, 1)$

d) $A(2, 2)$.

参考答案:

a) 0

b) 2

c) 2

d) 4

$$A(m, n) = \begin{cases} 2n & \text{if } m = 0 \\ 0 & \text{if } m \geq 1 \text{ and } n = 0 \\ 2 & \text{if } m \geq 1 \text{ and } n = 1 \\ A(m-1, A(m, n-1)) & \text{if } m \geq 1 \text{ and } n \geq 2 \end{cases}$$

6. 1

48. In how many ways can a photographer at a wedding arrange 6 people in a row from a group of 10 people, where the bride and the groom are among these 10 people, if

- a) the bride must be in the picture?
- b) both the bride and groom must be in the picture?
- c) exactly one of the bride and the groom is in the picture?

参考答案:

a) We first place the bride in any of the 6 positions. Then, from left to right in the remaining positions, we choose the other five people to be in the picture; this can be done in $9 \cdot 8 \cdot 7 \cdot 6 \cdot 5 = 15120$ ways. Therefore the answer is $6 \cdot 15120 = 90,720$.

b) We first place the bride in any of the 6 positions, and then place the groom in any of the 5 remaining positions. Then, from left to right in the remaining positions, we choose the other four people to be in the picture; this can be done in $8 \cdot 7 \cdot 6 \cdot 5 = 1680$ ways. Therefore the answer is $6 \cdot 5 \cdot 1680 = 50,400$.

c) From part (a) there are 90720 ways for the bride to be in the picture. There are (from part (b)) 50400 ways for both the bride and groom to be in the picture. Therefore there are $90720 - 50400 = 40320$ ways for just the bride to be in the picture. Symmetrically, there are 40320 ways for just the groom to be in the picture. Therefore the answer is $40320 + 40320 = 80,640$.

6. 1 52. How many bit strings of length 10 contain either five consecutive 0's or five consecutive 1's?

参考答案:

First we count the number of bit strings of length 10 that contain five consecutive 0's. We will base the count on where the string of five or more consecutive 0's starts.

(1) If it starts in the first bit, then the first five bits are all 0's, but there is free choice for the last five bits; therefore there are $2^5 = 32$ such strings.

(2) If it starts in the second bit, then the first bit must be a 1, the next five bits are all 0's, but there is free choice for the last four bits; therefore there are $2^4 = 16$ such strings.

(3) If it starts in the third bit, then the second bit must be a 1 but the first bit and the last three bits are arbitrary; therefore there are $2^4 = 16$ such strings.

(4) Similarly, there are 16 such strings that have the consecutive 0's starting in each of positions four, five, and six. This gives us a total of $32 + 5 \cdot 16 = 112$ strings that contain five consecutive 0's. Symmetrically, there are 112 strings that contain five consecutive 1's. Clearly there are exactly two strings that contain both (0000011111 and 1111100000). Therefore by the inclusion-exclusion principle, the answer is $112 + 112 - 2 = 222$.

6.2 14. How many ordered pairs of integers (a, b) are needed to guarantee that there are two ordered pairs (a_1, b_1) and (a_2, b_2) such that $a_1 \bmod 5 = a_2 \bmod 5$ and $b_1 \bmod 5 = b_2 \bmod 5$?

参考答案:

Working modulo 5 there are 25 pairs: $(0, 0), (0, 1), \dots, (4, 4)$. Thus we could have 25 ordered pairs of integers (a, b) such that no two of them were equal when reduced modulo 5. The pigeonhole principle, however, guarantees that if we have 26 such pairs, then at least two of them will have the same coordinates, modulo 5.

6.3:30. A professor writes 40 discrete mathematics true/false questions. Of the statements in these questions, 17 are true. If the questions can be positioned in any order, how many different answer keys are possible?

Solution: $C(40, 17)$

46. How many ways are there for a horse race with four horses to finish if ties are possible? [Note: Any number of the four horses may tie.]

We can solve this problem by breaking it down into cases depending on the number of ties. There are five cases.

- (1) If there are no ties, then there are clearly $P(4,4) = 24$ possible ways for the horses to finish.
- (2) Assume that there are two horses that tie, but the others have distinct finishes. There are $C(4,2) = 6$ ways to choose the horses to be tied; then there are $P(3,3) = 6$ ways to determine the order of finish for the three groups (the pair and the two single horses). Thus there are $6 \cdot 6 = 36$ ways for this to happen.
- (3) There might be two groups of two horses that are tied. There are $C(4,2) = 6$ ways to choose the winners (and the other two horses are the losers).
- (4) There might be a group of three horses all tied. There are $C(4,3) = 4$ ways to choose which these horses will be, and then two ways for the race to end (the tied horses win or they lose), so there are $4 \cdot 2 = 8$ possibilities.
- (5) There is only one way for all the horses to tie. Putting this all together, the answer is $24+36+6+8+1=75$.

14. How many solutions are there to the equation $x_1 + x_2 + x_3 + x_4 = 17$, where x_1, x_2, x_3 , and x_4 are nonnegative integers?

Solution: $C(4+17-1, 17) = C(20, 17) = C(20, 3) = 1140$

20. How many solutions are there to the inequality $x_1 + x_2 + x_3 \leq 11$, where x_1, x_2 , and x_3 are nonnegative integers? [Hint: Introduce an auxiliary variable x_4 such that $x_1 + x_2 + x_3 + x_4 = 11$.]

Solution:

We introduce the nonnegative slack variable x_4 , and our problem becomes the same as the problem of counting the number of nonnegative integer solutions to $x_1 + x_2 + x_3 + x_4 = 11$. By Theorem 2 the answer is $C(4 + 11 - 1, 11) = C(14, 11) = C(14, 3) = 364$.

36. How many strings with five or more characters can be formed from the letters in SEERESS?

Solution :

We need to calculate separately, using Theorem 3, the number of strings of length 5, 6, and 7.

(1) There are $7!/(3!3!1!) = 140$ strings of length 7.

(2) For strings of length 6, we can omit the R and form $6!/(3!3!) = 20$ string; omit an E and form $6!/(3!2!1!) = 60$ strings, or omit an S and also form 60 strings. This gives a total of 140 strings of length 6.

(3) For strings of length 5, we can omit two E's or two S's, each giving $5!/(3!1!1!) = 20$ strings; we can omit one E and one S ($5!/(2!2!1!) = 30$ strings); or we can omit the R and either an E or an S ($5!/(3!2!) = 10$ strings each). This gives a total of 90 strings of length 5.

$$140+140+90=370$$

For a grand total of 370 strings of length 5 or greater.

48. A shelf holds 12 books in a row. How many ways are there to choose five books so that no two adjacent books are chosen?

[Hint: Represent the books that are chosen by bars and the books not chosen by stars. Count the number of sequences of five bars and seven stars so that no two bars are adjacent.]

Solution:

There are 5 bars (chosen books), and therefore there are 6 places where the 7 stars (nonchosen books) can fit (before the first bar, between the first and second bars, ..., after the fifth bar). Each of the second through fifth of these slots must have at least one star in it, so that adjacent books are not chosen. Once we have placed these 4 stars, there are 3 stars left to be placed in 6 slots. The number of ways to do this is therefore $C(6+3-1, 3) = C(8, 3) = 56$.